Practice Set B

Class 12 Mathematics, Chapter 4: Determinants

Maximum marks: **54** **20** questions

Name _____ Date _____ Score _____ / 54

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1

8 marks

1 MCQ, NCERT Exemplar — 1 mark

There are two values of a which make the determinant $\Delta = \begin{vmatrix} 1 & -2 & 5 \\ 2 & a & -1 \\ 0 & 4 & 2a \end{vmatrix} = 86$. The sum of these two values is:

(A) 4 **(B)** 5 **(C)** -4 **(D)** 9

2 MCQ, CBSE SQP 2025-26 — 1 mark

If A is a square matrix of order 4 and $|\text{adj } A| = 27$, then $A\,(\text{adj } A)$ is equal to:

(A) 3 **(B)** 9 **(C)** $3I$ **(D)** $9I$

3 MCQ, CBSE SQP 2023-24 — 1 mark

Given that A is a square matrix of order 3 and $|A| = -2$, then $|\text{adj}\,(2A)|$ is equal to:

(A) -2^6 **(B)** $+4$ **(C)** -2^8 **(D)** 2^8

4 MCQ, NCERT Exemplar — 1 mark

The area of a triangle with vertices $(-3, 0)$, $(3, 0)$ and $(0, k)$ is 9 sq units. The value of k will be:

(A) 9 **(B)** 3 **(C)** -9 **(D)** 6

5 MCQ, Board-pattern — 1 mark

If A_{ij} is the cofactor of the element a_{ij} of the determinant $\Delta = \begin{vmatrix} 1 & 2 & -1 \\ 3 & 0 & 4 \\ -2 & 5 & 1 \end{vmatrix}$, then the value of $a_{11}A_{21} + a_{12}A_{22} + a_{13}A_{23}$ is:

(A) -57 **(B)** 0 **(C)** 57 **(D)** -4

Q No.	Question	Marks
6	Assertion-Reason, Board-pattern **Assertion (A):** If A is a square matrix of order 3 with $\lvert A\rvert = -2$, then $\lvert\text{adj}\,(\text{adj}\,A)\rvert = 16$. **Reason (R):** For a square matrix A of order n, $\lvert\text{adj}\,A\rvert = \lvert A\rvert^{n-1}$. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
7	Short answer, PYQ-style Find the values of k if the area of the triangle with vertices $(1, -1)$, $(-4, 2k)$ and $(-k, -5)$ is 24 sq units.	2 marks

Level 2

17 marks

8	Short answer, Board-pattern Find the adjoint and the inverse of $A = \begin{bmatrix} a & b \\ c & \dfrac{1+bc}{a} \end{bmatrix}$, where $a \neq 0$.	2 marks
9	Short answer, NCERT Exemplar Find A^{-1} if $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ and show that $A^{-1} = \dfrac{A^2 - 3I}{2}$.	3 marks
10	Short answer, Board-pattern Using determinants, find the equation of the line joining $A(2, -3)$ and $B(-1, 4)$. Hence find k if the point $C(k, -10)$ lies on this line.	2 marks
11	Short answer, CBSE 2023 If $A = \begin{bmatrix} 3 & 2 \\ 5 & -7 \end{bmatrix}$, then find A^{-1} and use it to solve the following system of equations: $3x + 5y = 11$, $2x - 7y = -3$.	3 marks
12	Short answer, Board-pattern Using the matrix method, find the value of k for which the system of equations $x + 2y = 3$, $3x + ky = 5$ has no solution.	2 marks
13	Short answer, PYQ-style If $A = \begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$, show that $A'A^{-1} = \begin{bmatrix} \cos 2x & -\sin 2x \\ \sin 2x & \cos 2x \end{bmatrix}$.	2 marks
14	Short answer, Board-pattern If $A^{-1} = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, find $(AB)^{-1}$.	3 marks

Q No.	Question	Marks

Level 3 — 29 marks

15 Long answer, PYQ-style — 5 marks

A school wants to award its students for the values of Honesty, Regularity and Hard work with a total cash award of ₹6,000. Three times the award money for Hard work added to that given for Honesty amounts to ₹11,000. The award money given for Honesty and Hard work together is double the one given for Regularity. Represent the situation algebraically and find the award money for each value, using the matrix method.

16 Long answer, CBSE 2025 — 5 marks

Given $A = \begin{bmatrix} -4 & 4 & 4 \\ -7 & 1 & 3 \\ 5 & -3 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -1 & 1 \\ 1 & -2 & -2 \\ 2 & 1 & 3 \end{bmatrix}$, find AB. Hence, solve the system of linear equations: $x - y + z = 4$, $x - 2y - 2z = 9$, $2x + y + 3z = 1$.

17 Long answer, NCERT Misc. Ex. Q7 — 5 marks

Solve the following system of equations by the matrix method: $\frac{2}{x} + \frac{3}{y} + \frac{10}{z} = 4$, $\frac{4}{x} - \frac{6}{y} + \frac{5}{z} = 1$, $\frac{6}{x} + \frac{9}{y} - \frac{20}{z} = 2$.

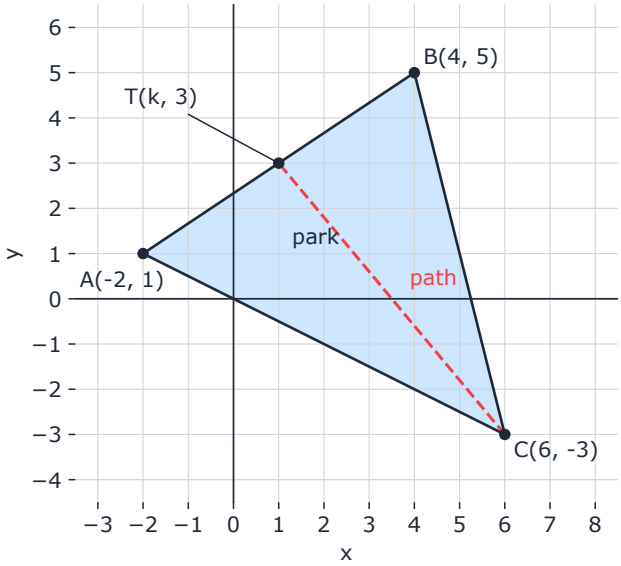
18 Long answer, Board-pattern — 5 marks

Ritu invests a total of ₹10,000 in three schemes P, Q and R, which pay simple interest at 5%, 8% and 9% per annum respectively. Her total interest for one year is ₹770. The amount invested in scheme Q is equal to the sum of the amounts invested in P and R. Using the matrix method, find the amount invested in each scheme.

19 Long answer, PYQ-style — 5 marks

If $A = \begin{bmatrix} 2 & 1 & -3 \\ 3 & 2 & 1 \\ 1 & 2 & -1 \end{bmatrix}$, find A^{-1}. Using A^{-1}, solve the system of equations: $2x + 3y + z = 4$, $x + 2y + 2z = 6$, $-3x + y - z = -10$.

Q No.	Question	Marks
20	Case study, Board-pattern **Case Study:** A town park is in the shape of a triangle with corners $A(-2, 1)$, $B(4, 5)$ and $C(6, -3)$ on a map grid (1 unit $= 10$ m). A water tap $T(k, 3)$ is fixed on the fence AB, and a straight path is to be laid from the gate at C to the tap T.	4 marks



(i) Using determinants, find the area of the park in square metres. 1 mark

(ii) Using determinants, find the value of k. 1 mark

(iii) Using determinants, find the equation of the path CT. 2 marks

Answer key

Chapter 4: Determinants, Practice Set B. Full step-wise solutions are in the solutions sheet.

Level 1

1	(C) -4
2	(C) $3I$
3	(D) 2^8
4	(B) 3
5	(B) 0
6	(A) Both A and R are true and R is the correct explanation of A.
7	$k = 3$ or $k = -\dfrac{9}{2}$

Level 2

8	$\operatorname{adj} A = A^{-1} = \begin{bmatrix} \dfrac{1+bc}{a} & -b \\ -c & a \end{bmatrix}$
9	$A^{-1} = \dfrac{1}{2}\begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix} = \dfrac{A^2 - 3I}{2}$
10	$7x + 3y - 5 = 0$; $k = 5$
11	$A^{-1} = -\dfrac{1}{31}\begin{bmatrix} -7 & -2 \\ -5 & 3 \end{bmatrix}$; $x = 2$, $y = 1$
12	$k = 6$
13	$A'A^{-1} = \begin{bmatrix} \cos 2x & -\sin 2x \\ \sin 2x & \cos 2x \end{bmatrix}$. Shown.
14	$(AB)^{-1} = -\dfrac{1}{2}\begin{bmatrix} 22 & -8 \\ -14 & 5 \end{bmatrix} = \begin{bmatrix} -11 & 4 \\ 7 & -\frac{5}{2} \end{bmatrix}$

Level 3

15	Honesty ₹500, Regularity ₹2,000, Hard work ₹3,500
16	$AB = 8I$; $x = 3$, $y = -2$, $z = -1$
17	$x = 2$, $y = 3$, $z = 5$
18	P: ₹2,000, Q: ₹5,000, R: ₹3,000
19	$A^{-1} = -\dfrac{1}{16}\begin{bmatrix} -4 & -5 & 7 \\ 4 & 1 & -11 \\ 4 & -3 & 1 \end{bmatrix}$; $x = 2$, $y = -1$, $z = 3$
20	(i) 2800 m^2; (ii) $k = 1$; (iii) $6x + 5y - 21 = 0$